



EduVision

Smarter Data
Better Decisions
Greater Impact

Higher Education Performance Analytics

Project Documentation & Delivery

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Infosys Springboard Internship 7.0 – Batch 2 | 2026



Project Overview

Project Purpose



To transform higher education data into meaningful visual insights that help users understand university performance, student outcomes, research impact, and global education trends.

Problem



Higher education data is often large, scattered, and difficult to interpret. Without interactive visualizations, comparing universities, countries, research performance, and student-related metrics becomes time-consuming.

What the Dashboard Provides



Interactive visual analytics for higher education data



University performance & ranking analysis



Research, publications & citation insights



Student analytics & performance indicators



Country-wise comparison of education metrics



Filters and interactive exploration for deeper analysis

Turning complex higher education data into clear, interactive, and actionable insights.

Project Objective

01 ANALYZE UNIVERSITY PERFORMANCE

Evaluate universities using ranking and academic indicators.

02 UNDERSTAND RESEARCH PERFORMANCE

Analyze publications, citations, research productivity, and research impact.

03 STUDY STUDENT INDICATORS

Explore student-related metrics, international representation, and faculty/student characteristics.

04 ENABLE INSTITUTIONAL COMPARISON

Compare universities across countries, regions, and performance indicators.

05 IDENTIFY GLOBAL PATTERNS

Discover trends and differences in higher education performance across countries and institutions.

Key Goal | Convert complex educational data into clear, interactive and decision-supporting visual insights.

Dataset Overview



QS World University Rankings 2026

KEY INDICATORS / METRICS

- 🏆 University rankings
- ★ Academic reputation
- 👥 Employer reputation
- 👨‍🎓 Faculty / student indicators
- 🌐 Internationalization
- 🔍 Research metrics

ABOUT THE SOURCE



QS Overview

QS World University Rankings 2026 includes over **1,504 universities** across **100+ locations**.



THE World University Rankings 2026

KEY INDICATORS / METRICS

- 📊 Overall performance
- 📖 Teaching
- 🔬 Research
- ” Student-staff ratio
- 🌐 International outlook
- 📈 Industry impact

ABOUT THE SOURCE



THE Overview

THE World University Rankings 2026 ranks **2,191 institutions** from **115 countries/territories**.

Data Categories Used



University Information



Country / Region



Ranking Metrics



Academic Indicators



Research Indicators



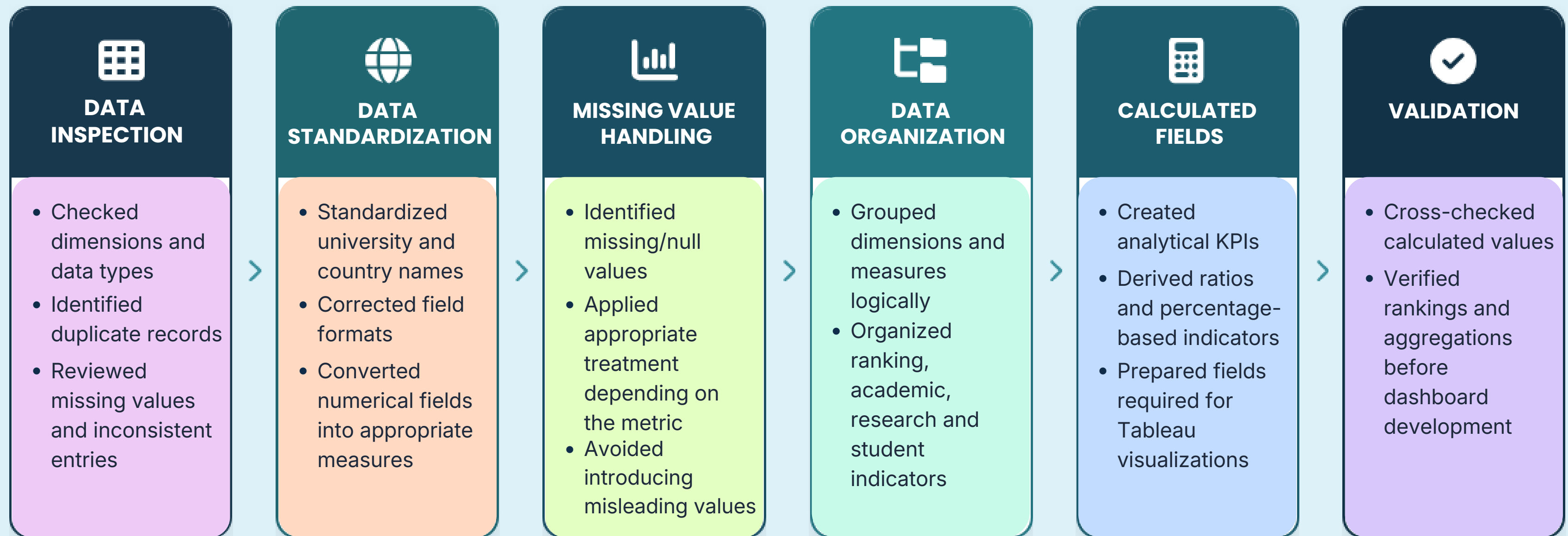
Student / Faculty



Internationalization

Data Preparation & Cleaning

Ensuring Data Integrity for Reliable University Analytics



Result: A robust, consistent dataset, ready for creating insightful and decision-supporting visual analysis.

EduVision Methodology



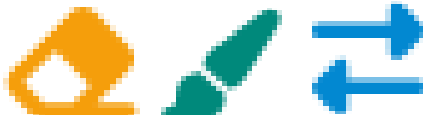
Step 1 — Data Collection

QS 2026 + THE 2026 datasets



Step 2 — Data Cleaning

Missing values • duplicates
• formatting • standardization



Step 3 — KPI Calculation

Ranking • Research
• Academic • Student
• Internationalization metrics



Step 4 — Exploratory Analysis

Patterns • comparisons • relationships • distributions



Step 5 — Visualization

Interactive Tableau charts, KPIs, filters and dashboards



Step 6 — Insight Generation

Identify high-performing institutions, countries, research patterns and student trends



Analytical Approach: • Descriptive analytics  • Comparative analysis  • Exploratory visual analytics 

Tech Stack

Category	Technologies & Tools	Process Stage	Key Outputs
DATA SOURCE	CSV, Excel	Data Acquisition	Raw Datasets (Files)
DATA PROCESSING & KPI ENGINEERING	Python, Jupyter, Pandas, Numpy	EDA & KPI Engineering	Clean Data & Metrics
DATA VISUALIZATION	Tableau	Visuals & Dashboards	Interactive Dashboards
DOCUMENTATION & PRESENTATION	Canva, PowerPoint	Reporting	Reports
PROJECT DELIVERY	GitHub	Code Management	Repository

KPI Engineering

Global Ranking Score – Overall ranking performance of the university
Average of normalized QS and THE rank scores

Research Impact Score – Measures the influence and impact of the university's research output.
Average of Citations per Faculty, THE Research Quality, Research Impact

Faculty-to-Student Ratio – 1 / Students-to-Staff Ratio

International Student Percentage – International Students × 100

Academic Reputation Score – Reflects the perceived academic quality and reputation of the university.
QS Academic Reputation Score

Research Productivity Index – Avg of Research Quality' Research Environment, Citations per Faculty, International Research Network, Research Impact

KPIs: Metrics that summarize the key performance areas of universities for meaningful comparison.

Dashboard 1 – University Overview

Provides a global macro-level view of higher education institutions based on top performance metrics, geographical spread, and overall QS score distribution.

KPI Metrics:

- **Total Universities: 535**
- **Avg QS Score: 46.84**
- **Avg THE Score: 41.15**
- **Avg Research Impact: 43.45**

Key Findings:

- Top Performers: Imperial College London (99.40), Univ. of Cambridge (97.90), and Univ. of Oxford (97.20) lead globally in QS Overall Scores.
- Scale Concentration: Large (49.16%) and Medium (23.55%) institutions make up over 72% of all analyzed universities.
- Reputation Correlation: Academic Reputation Score exhibits a strong direct correlation with QS Overall Performance.

Filters

Country

(All) ▾

Region

(All) ▾

Size

(All) ▾

Status

(All) ▾

🔄 Reset Filters

Higher Education Performance Dashboard

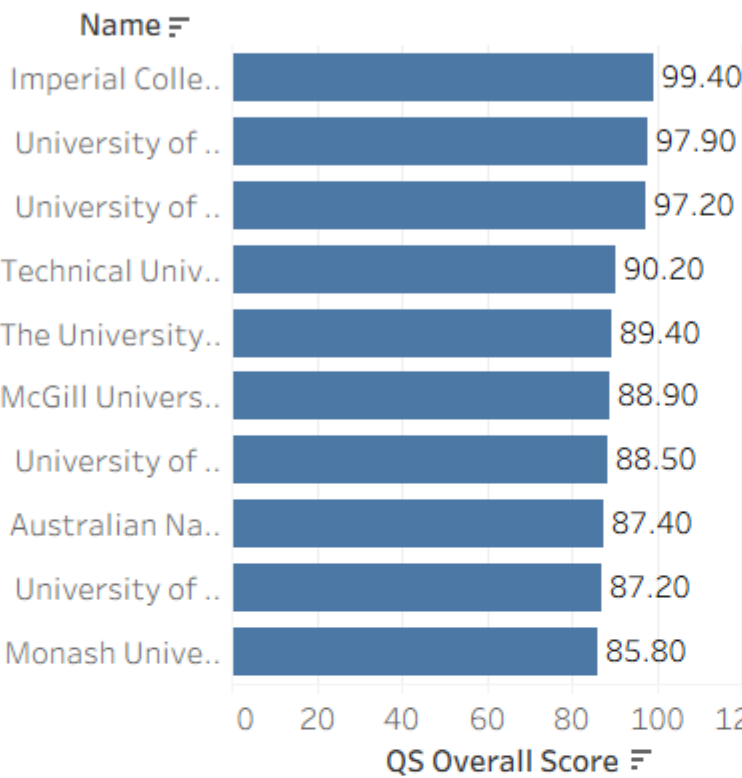
Total Universities
535

Average QS Score
46.84

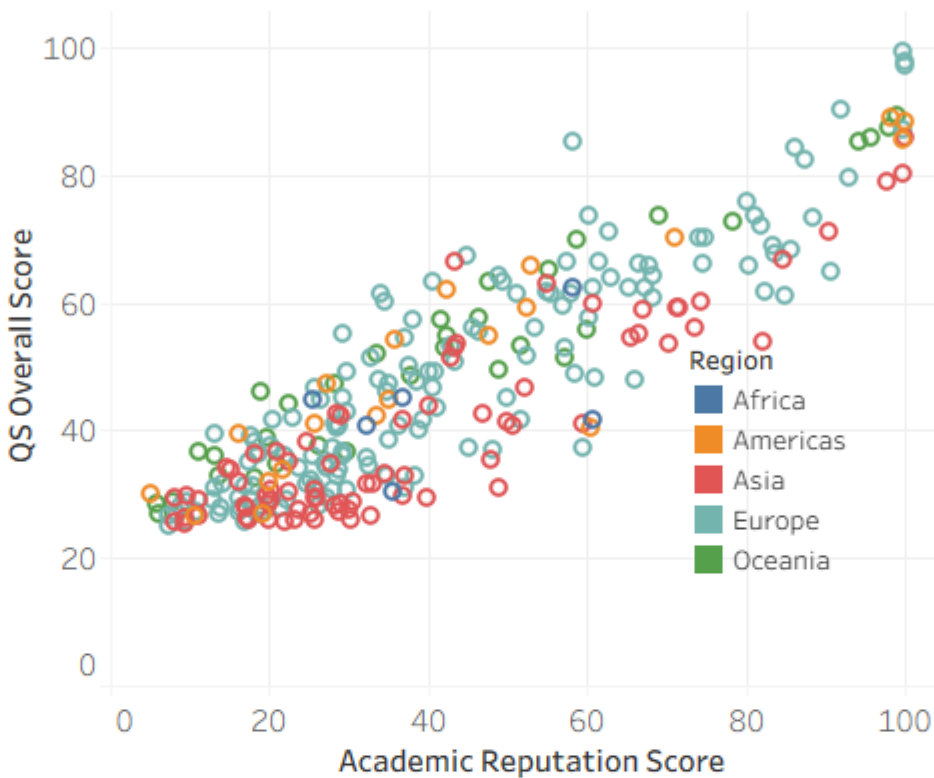
Average THE Score
41.15

Avg Research Impact
43.45

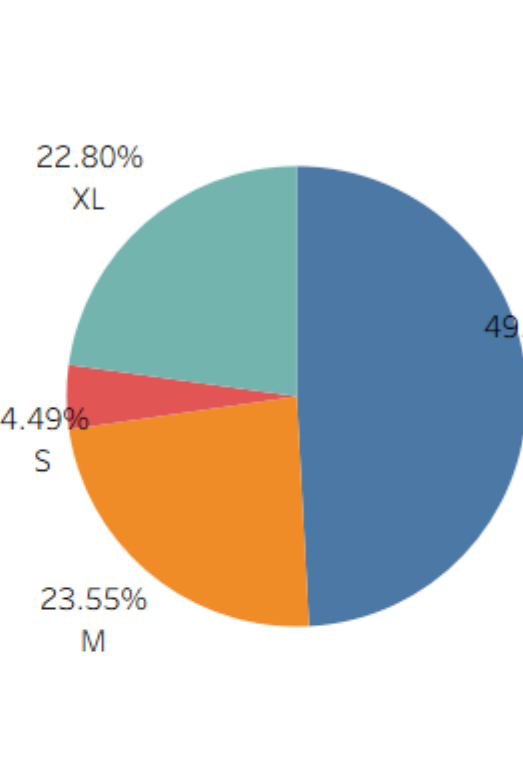
Top 10 Universities



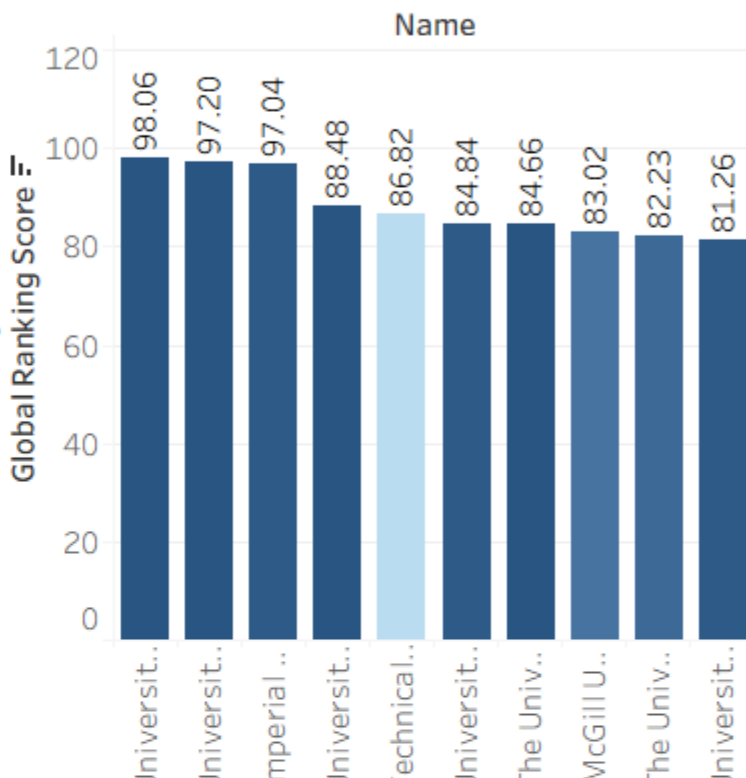
Academic Reputation Analysis



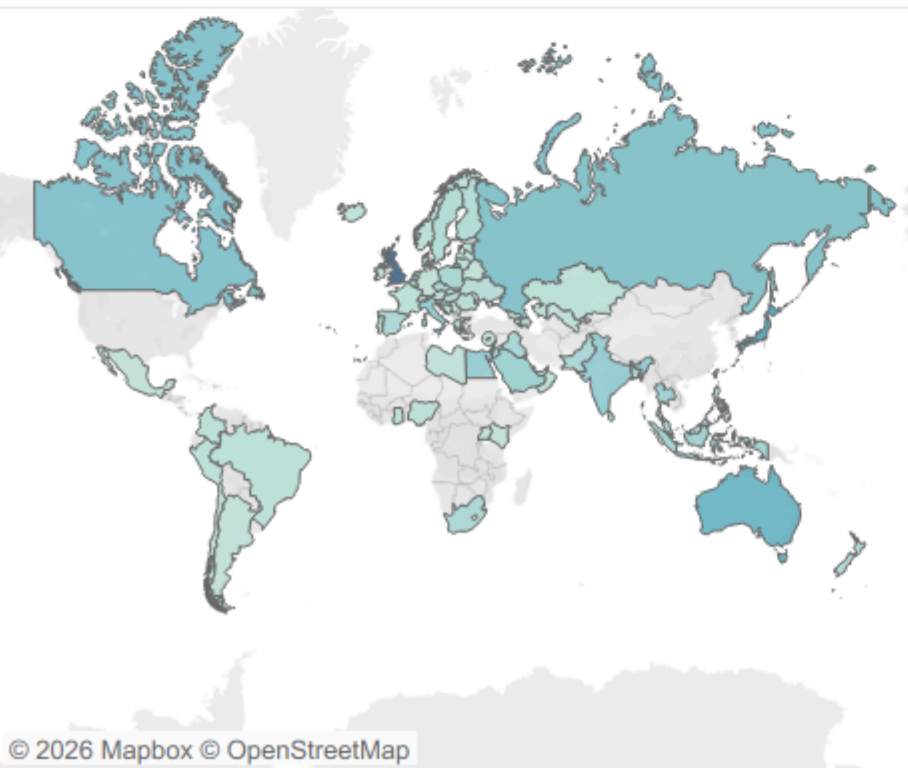
Universities by Size



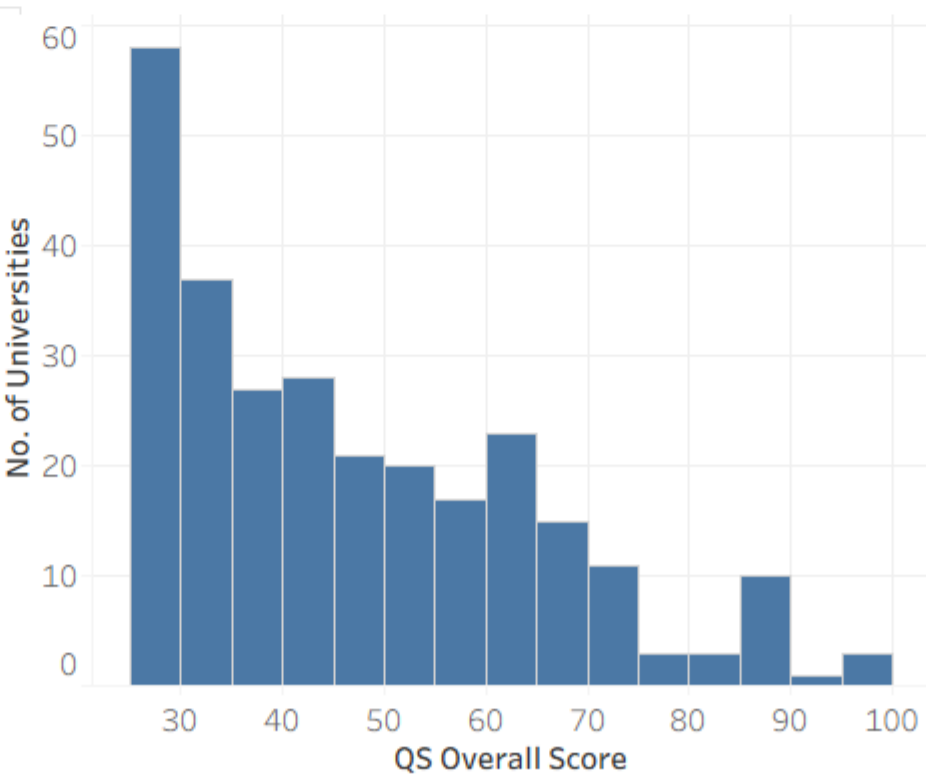
University Performance Trends



Global University Distribution



QS Overall Score Distribution



Institutional Comparison

Name	QS Overall Score	THE Overall Sco..	Academic Repu..	Research Impa..
Imperial Colle..	99.40	94.69	99.60	96.35
McGill Univer..	88.90	77.13	98.10	80.57
Technical Uni..	90.20	83.45	92.00	91.15
The Universit..	89.40	75.06	98.90	87.20
The Universit..	85.80	83.52	100.00	80.28
University of ..	85.50	77.03	99.60	82.50
University of ..	97.20	97.21	100.00	96.20
University of ..	87.20	82.49	99.60	80.82
University of ..	97.90	98.21	100.00	97.18
University of ..	88.50	88.45	100.00	87.75

Dashboard 2 – Research Analytics

Evaluates institutional research strength through publication output, citations per faculty, research impact, and quality scores.

KPI Metrics:

- **Avg Citations per Faculty: 30.11**
- **Avg Research Quality: 58.84**
- **Avg Research Productivity: 42.81**
- **Avg Research Impact: 43.45**

Key Findings:

- **Productivity Leaders:** Al-Ahliyya Amman Univ (56.35) and Alfaisal Univ (46.60) score highest on the Research Productivity Index.
- **Citation Excellence:** Institutions like Anna University, Tel Aviv University, and Univ. of Auckland hold near-perfect citation scores (~99.80).
- **Efficiency Link:** Strong linear relationship exists between Research Productivity Index and total Research Impact.

Filters

Country

(All) ▾

Region

(All) ▾

Size

(All) ▾

Status

(All) ▾

🔄 Reset Filters

Higher Education Performance Dashboard

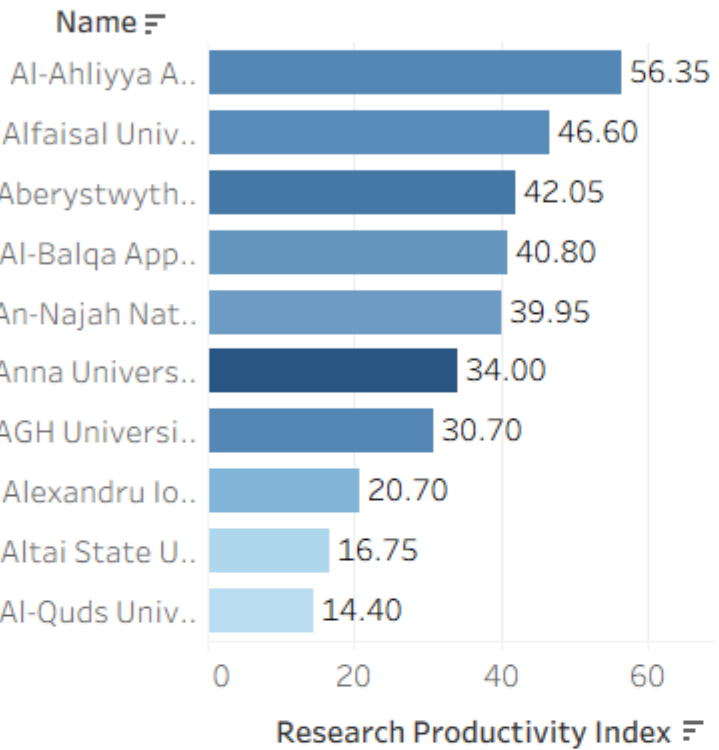
Avg Citations
30.11

Avg Research Quality
58.84

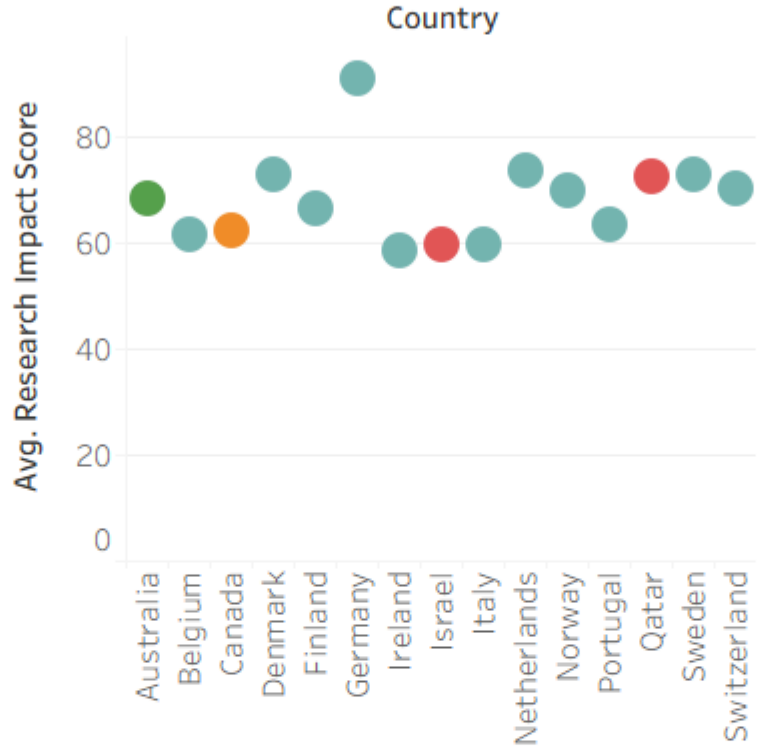
Avg Rsrch Productivity
42.81

Avg Research Impact
43.45

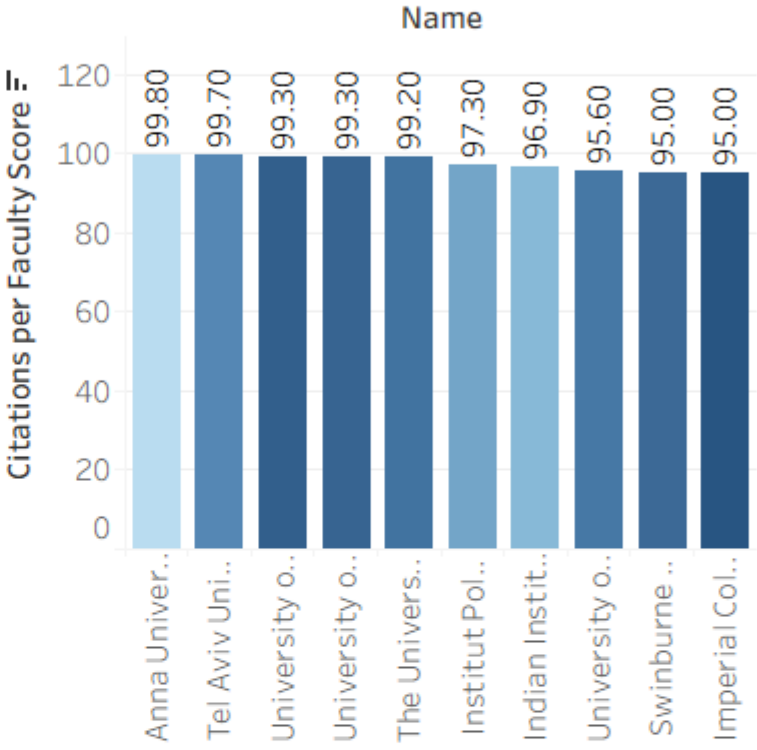
Top Research Institutions



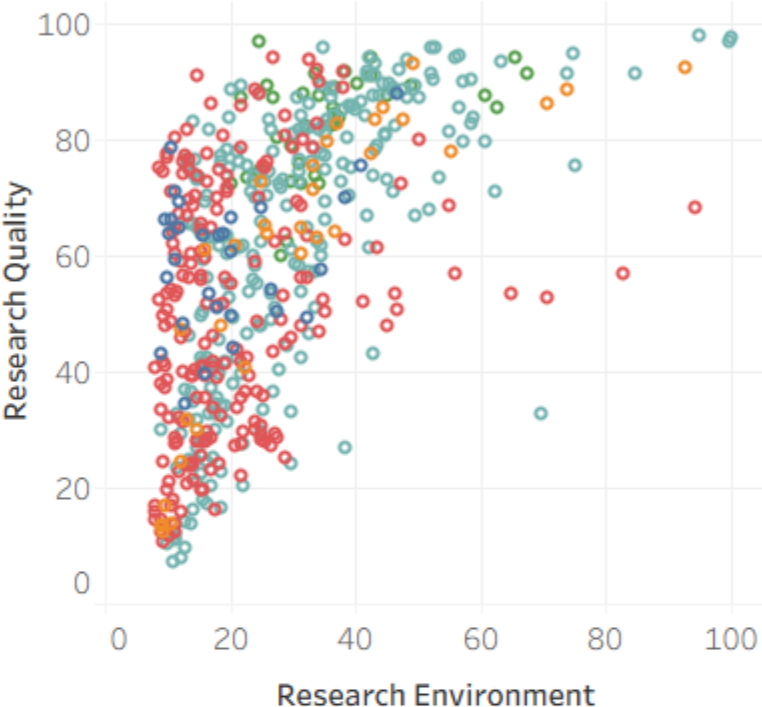
Top Countries by Research Impact



Citation Performance



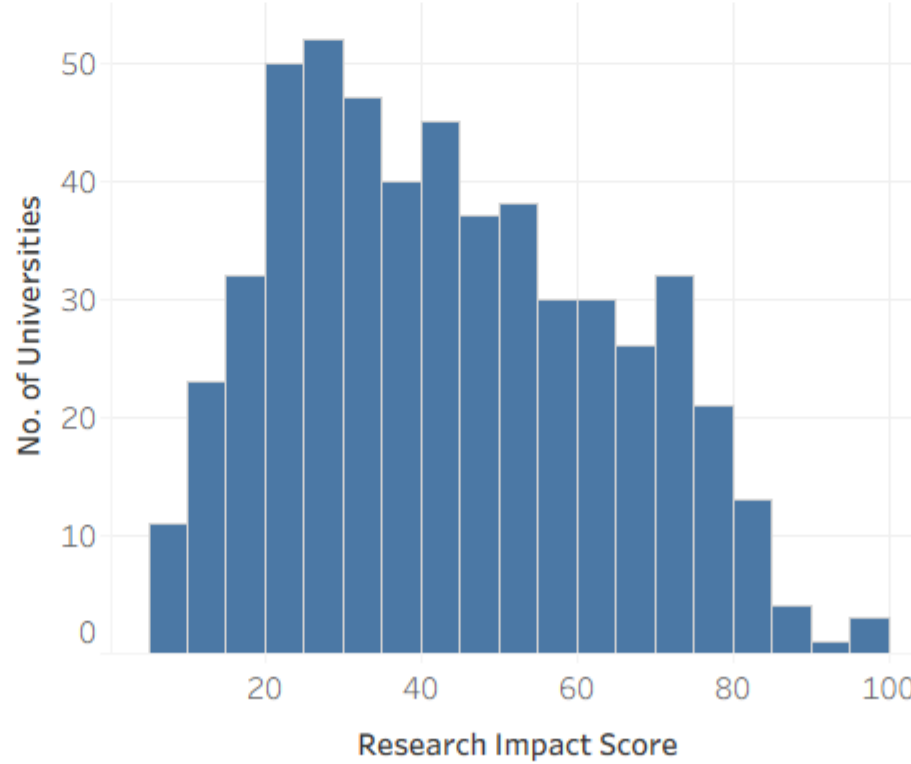
Research Quality vs Research Environment



Research Productivity Analysis



Research Impact Score Distribution



Research Impact Comparison

Name	Citations pe..	Research En..	Research Qu..	Internation..
Aberystwyth Univer..	23.90	22.20	61.90	63.30
AGH University of Kr..	12.70	21.60	39.80	75.60
Al-Ahliyya Amman U..	6.40	24.70	88.00	32.70
Al-Balqa Applied Uni..	5.80	16.90	64.70	46.00
Al-Quds University	2.80	11.00	17.80	15.00
Alexandru Ioan Cuza..	5.80	12.20	29.20	49.10
Alfaisal University	21.80	19.50	73.70	28.10
Altai State University	1.90	16.40	17.10	18.30
An-Najah National U..	4.30	13.00	66.90	39.00
Anna University	99.80	16.20	51.80	47.30

Dashboard 3 – Student Analytics

Focuses on student demographics, internationalization ratios, regional student distribution, and faculty-to-student capacity.

KPI Metrics:

- **Total Students: 12.84 Million**
- **International Student %: 16.50%**
- **Faculty/Student Ratio : 31.35**
- **Female/Male Ratio: 2.062**

Key Findings:

- Key Insights:
- Global Attraction: Ajman University (81%) and Islamic University (77%) lead in international student percentage.
- Regional Volume: United Kingdom, Russia, Egypt, and South Africa represent the largest student population blocks.
- Capacity Constraints: A majority of universities score low (<30) on Faculty-to-Student Ratio, signaling potential classroom overcrowding.

Filters

Country

(All)

Region

(All)

Size

(All)

Status

(All)

🔄 Reset Filters

Higher Education Performance Dashboard

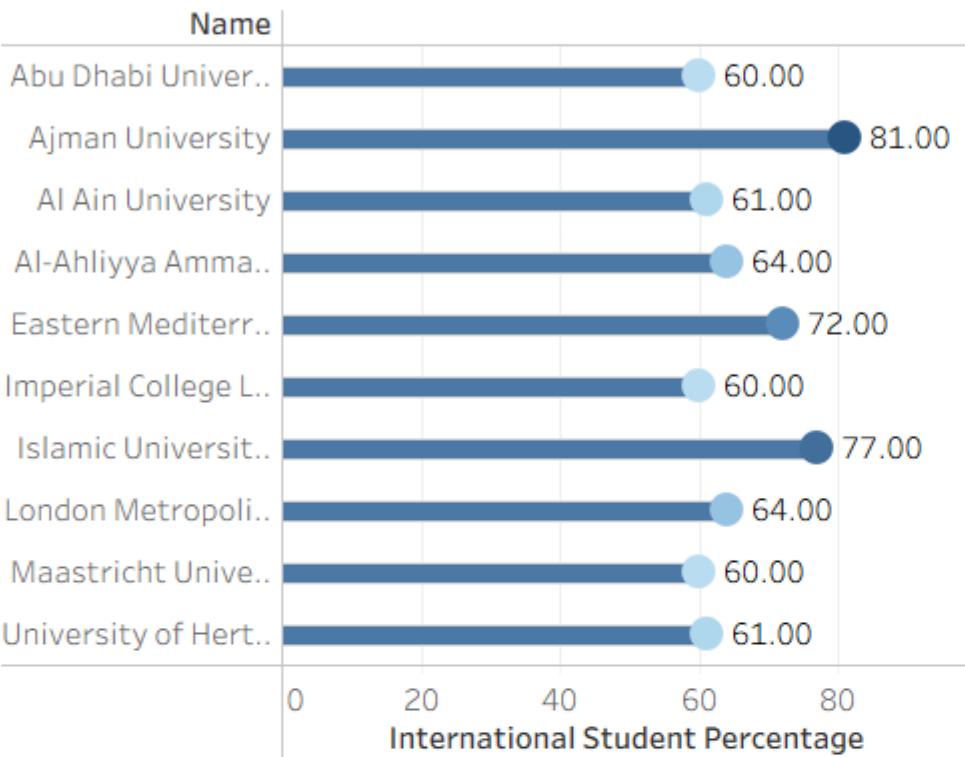
Total Students
12,844,047

International Student%
16.50

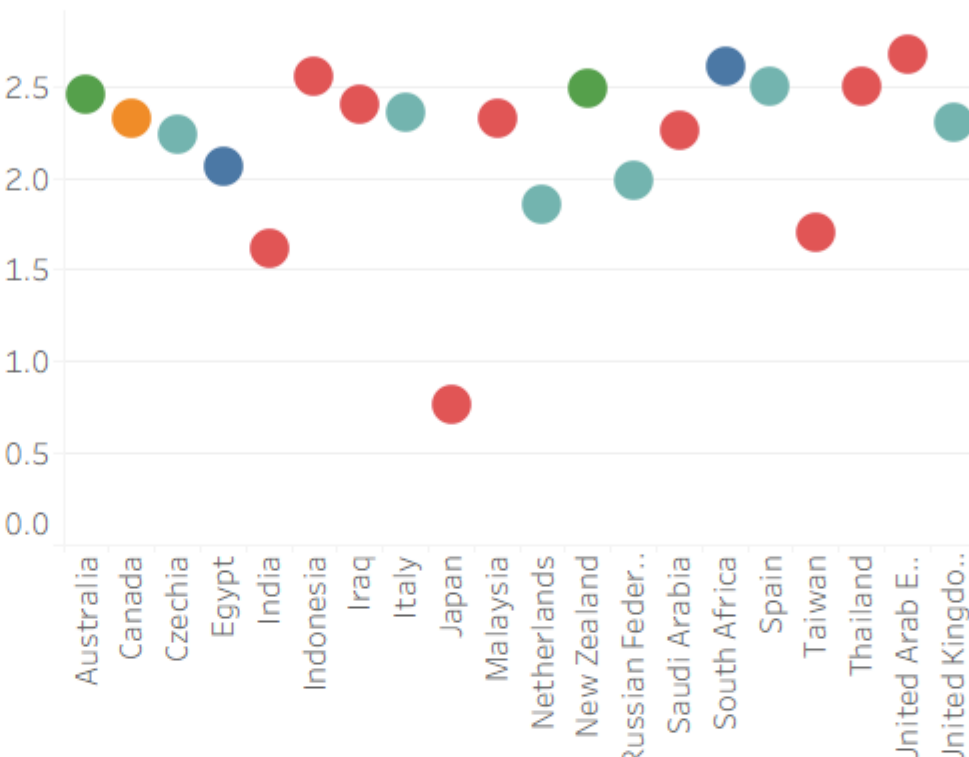
Faculty/Student Ratio
31.35

Female/Male Ratio
2.062

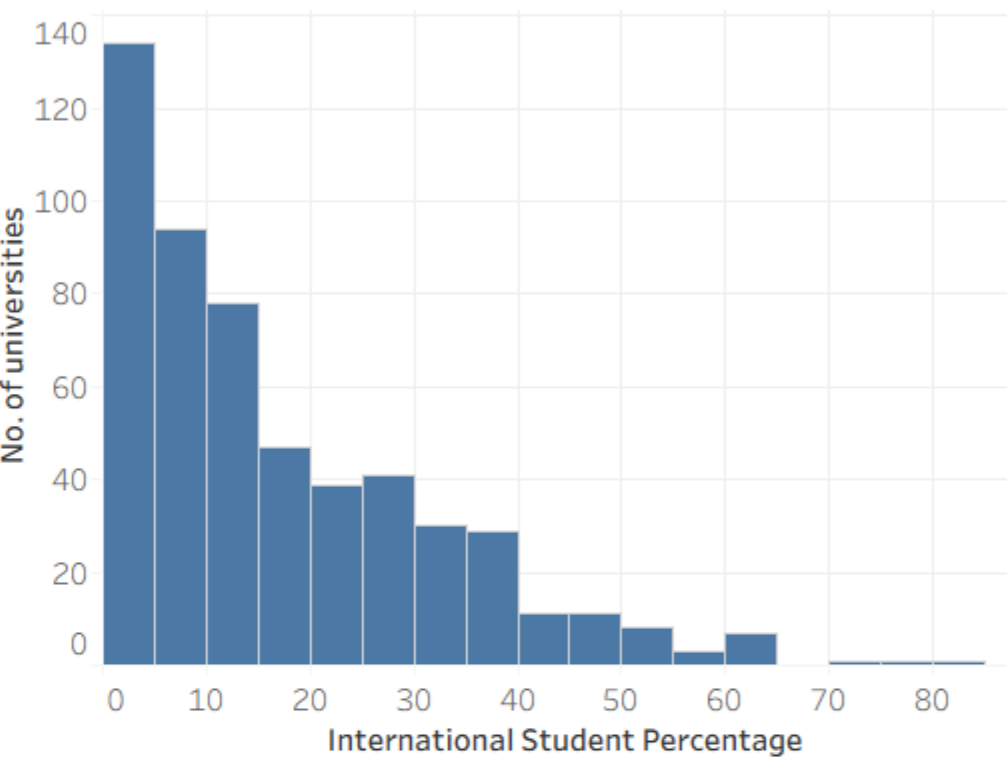
International Student Analysis



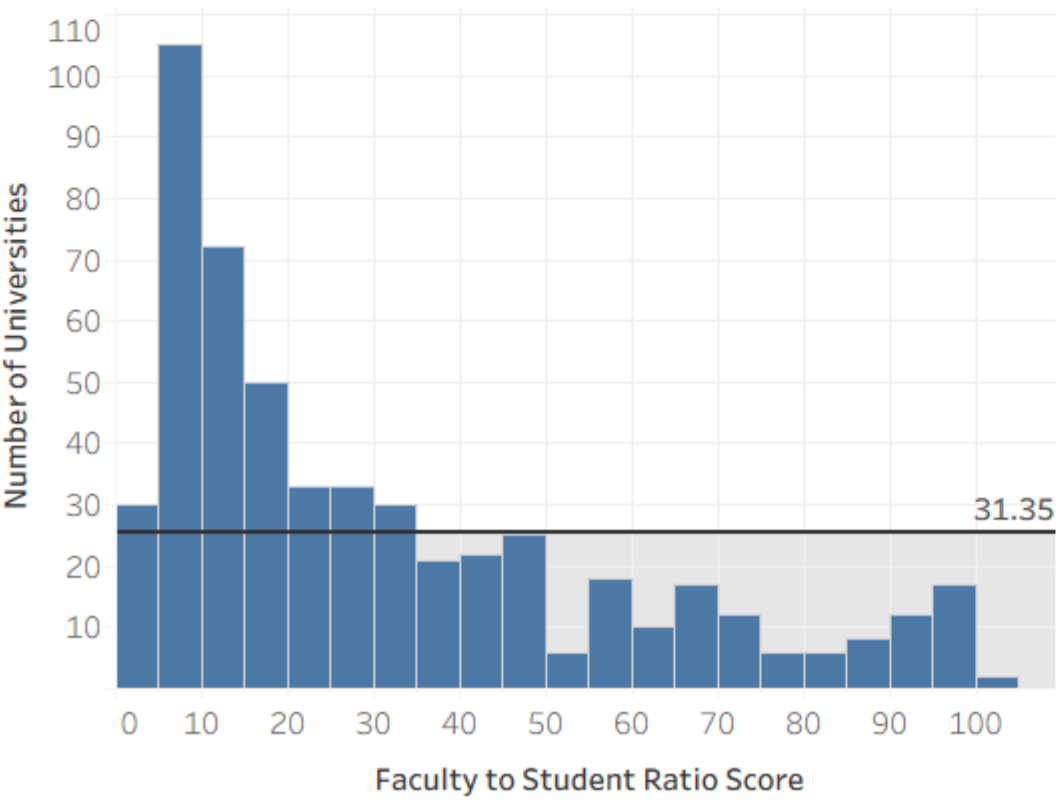
Top 20 Countries by Female-to-Male Ratio



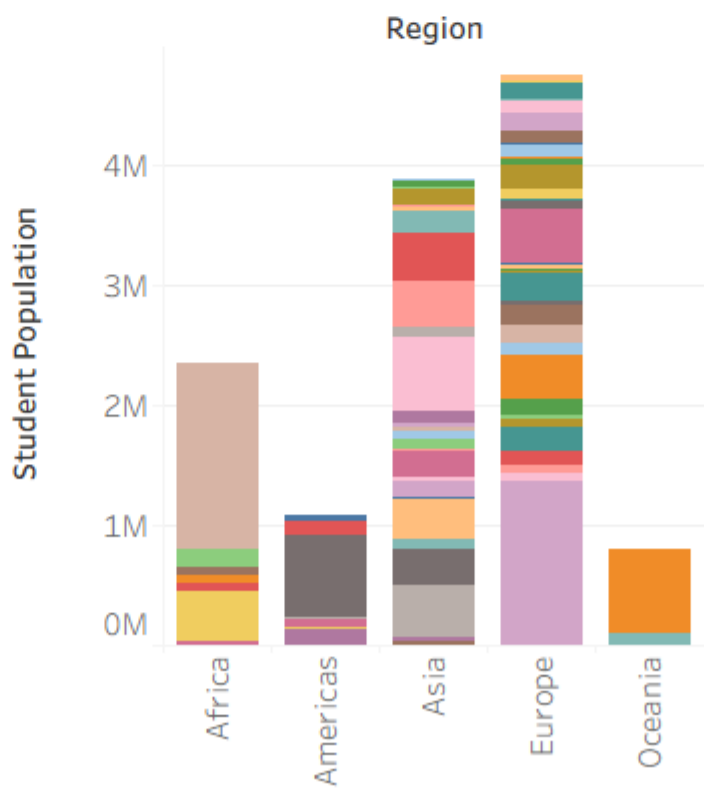
International Student Percentage Distribution



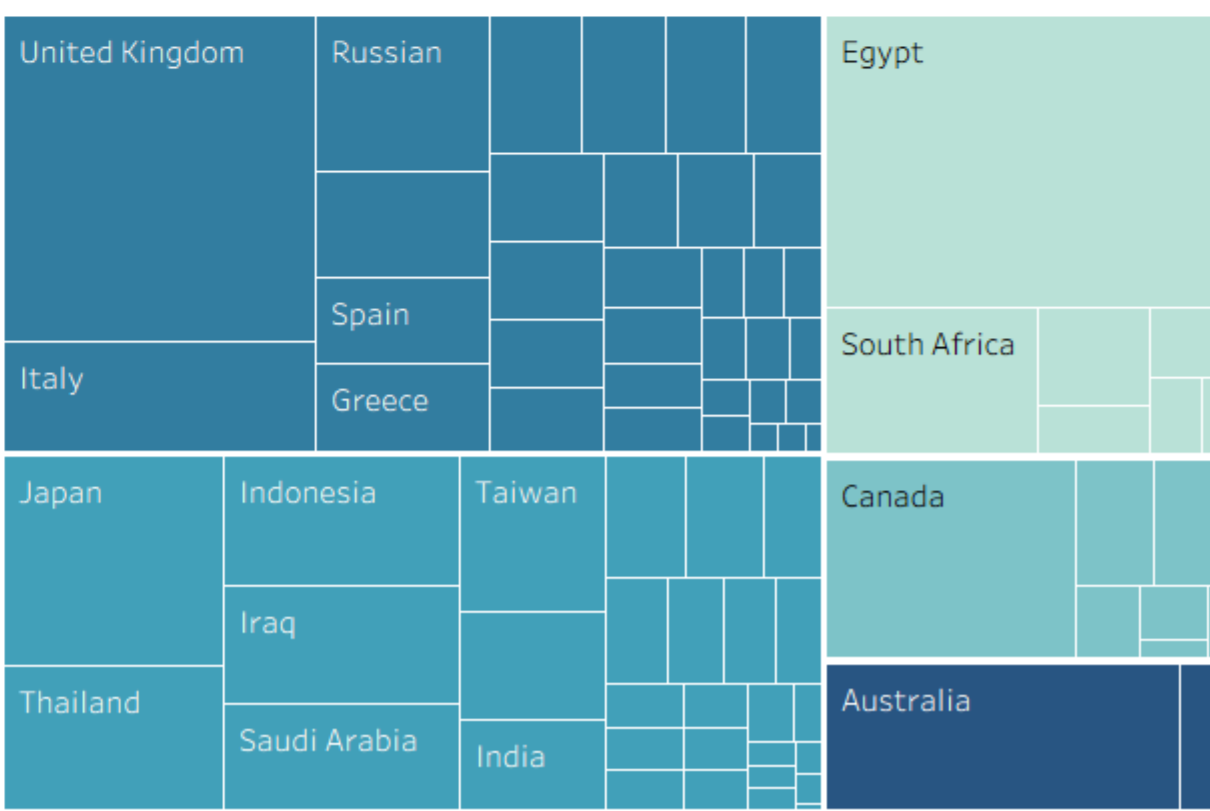
Faculty-to-Student Ratio Score Distribution



Student Population by Region



Student Distribution



Dashboard 4 – Country Comparison

Compares country-level performance, regional institution concentration, global score distributions, and reputation gaps.

KPI Metrics:

- **Avg Global Score: 48.90**
- **Avg Research Impact: 43.45**
- **Avg Academic Reputation: 27.12**
- **Total Universities: 535**

Key Findings:

- National Leaders: Germany, France, Sweden, and Qatar post the highest average Global Ranking Scores (>60).
- Regional Dominance: Europe (44.11%) and Asia (37.38%) combined host more than 81% of all top-ranked universities.
- Reputation vs. Impact Gap: Global Academic Reputation (27.12) trails average Research Impact (43.45), indicating strong research outputs despite lower global brand awareness.

Higher Education Performance Dashboard

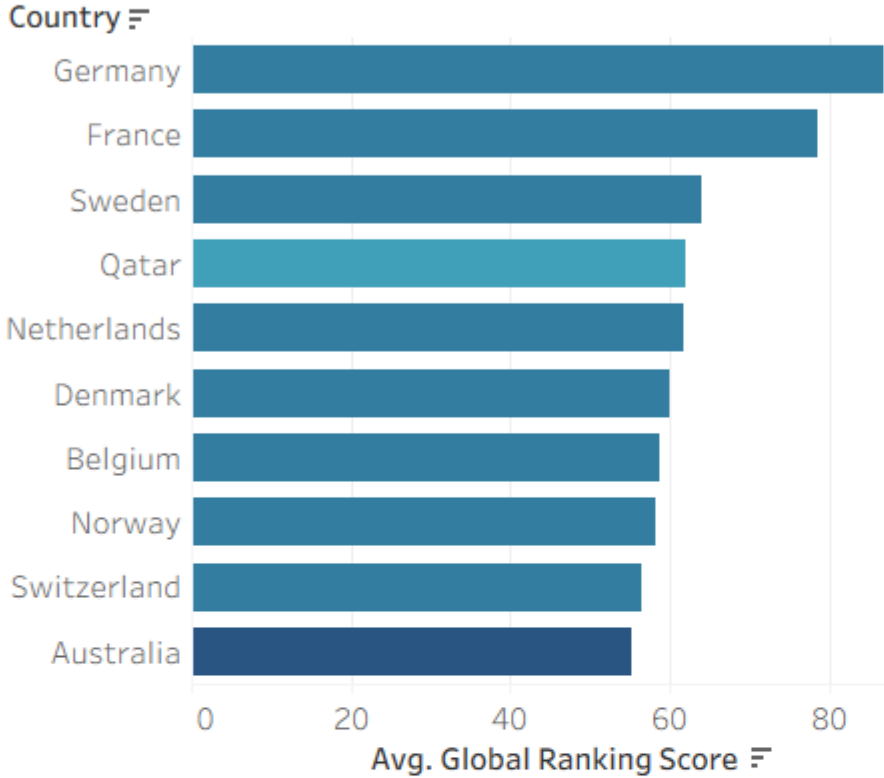
Avg Global Score
48.90

Avg Research Impact
43.45

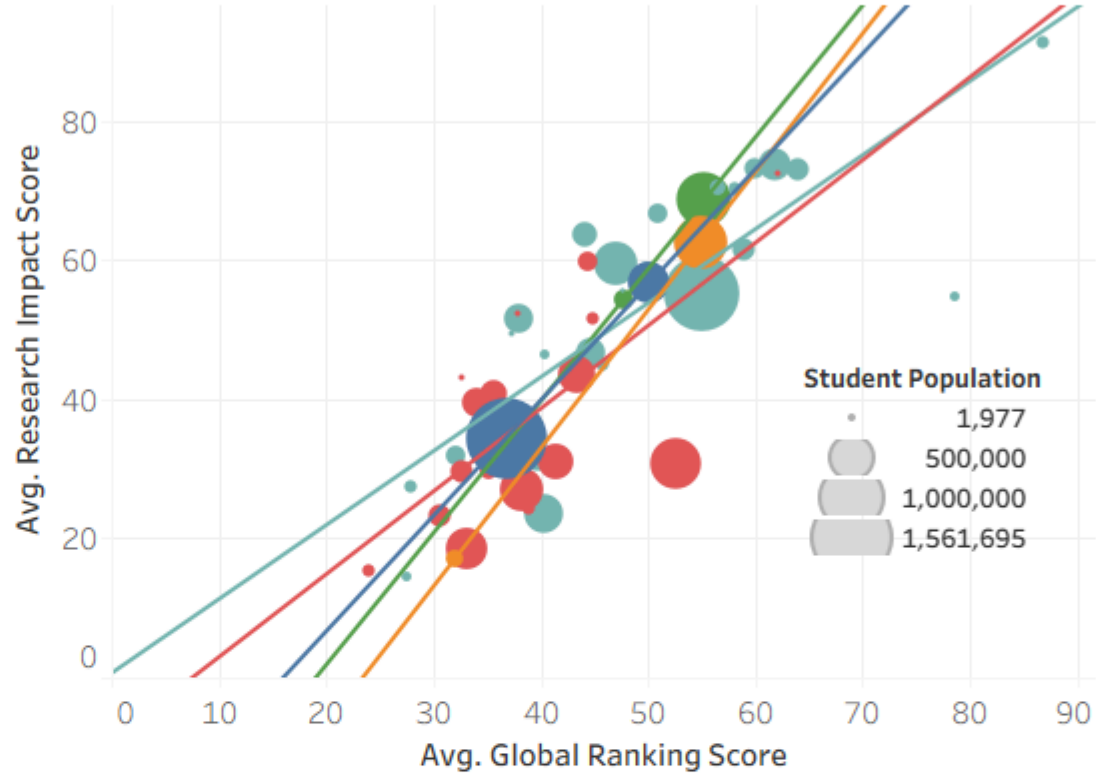
Avg Academic Reputation
27.12

Total Universities
535

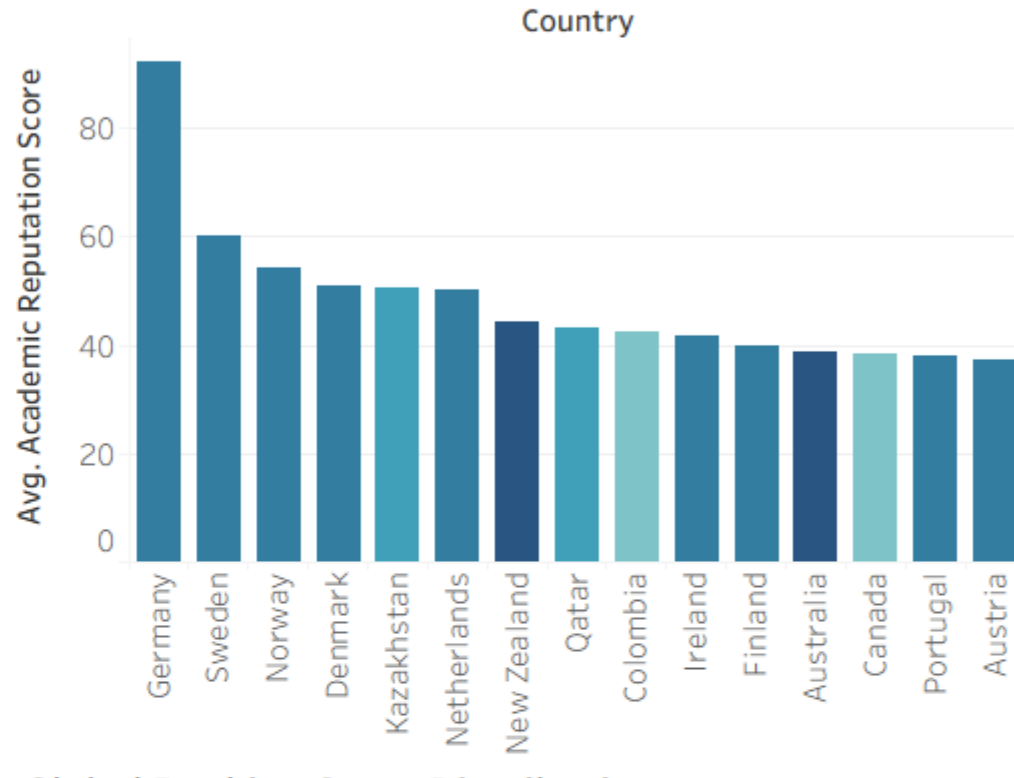
Country Ranking



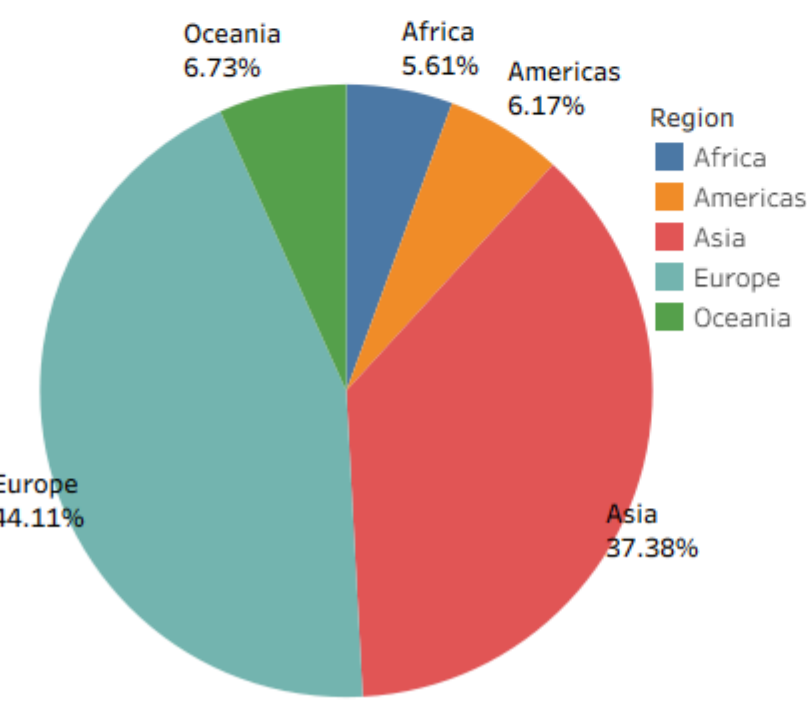
Research Impact vs Global Ranking



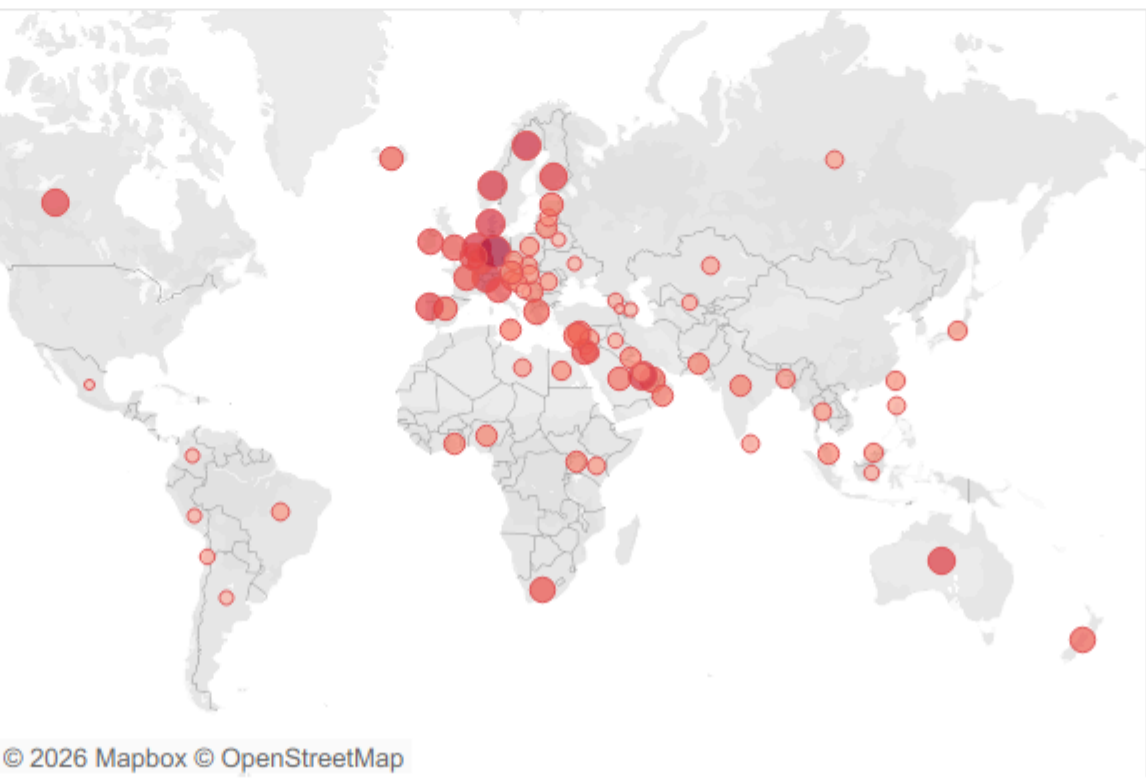
Academic Reputation by Country



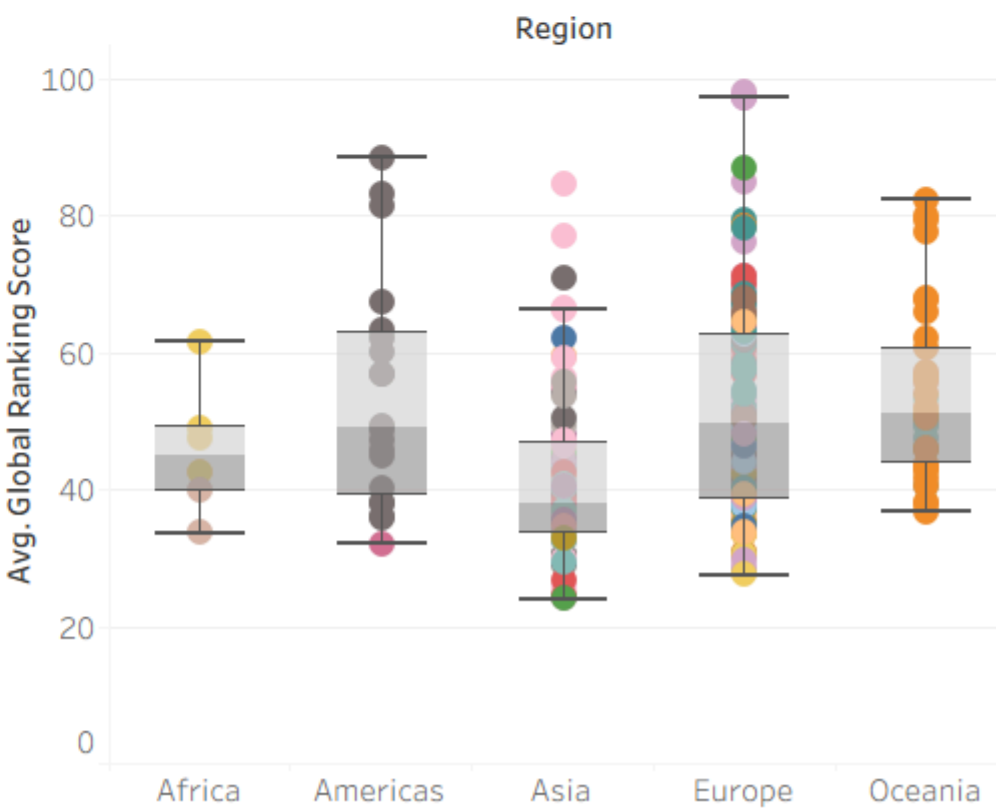
University Distribution



Research Impact by Country

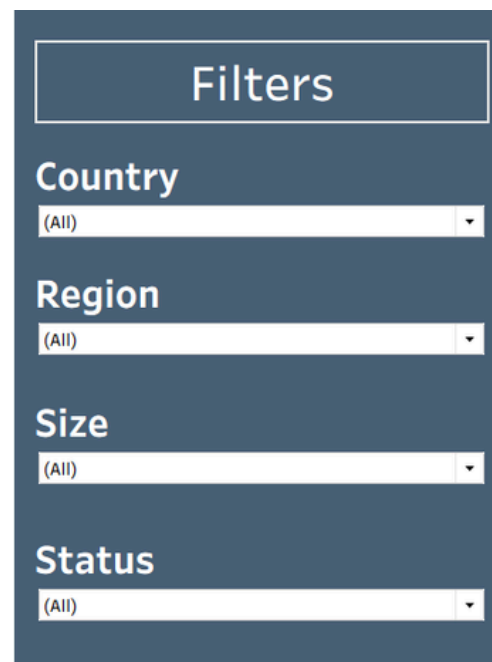


Global Ranking Score Distribution



Filters and Interactivity

1. Global Filters Panel (Sidebar Controls) ▼

A sidebar panel titled "Filters" with four dropdown menus: "Country" (All), "Region" (All), "Size" (All), and "Status" (All).

For real-time data slice-and-dice.

 **Country Dropdown:** Filters the entire dashboard suite by specific selected country.

 **Region Dropdown:** Categorizes analysis across continents (Europe, Asia, Americas, Africa, Oceania).

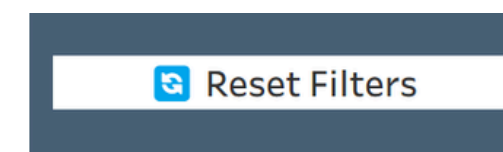
 **Size Dropdown:** Filters institutions based on enrollment scale (S, M, L, XL).

 **Status Dropdown:** Distinguishes between Public & Private university operational structures.

2. Reset Filters Action Button

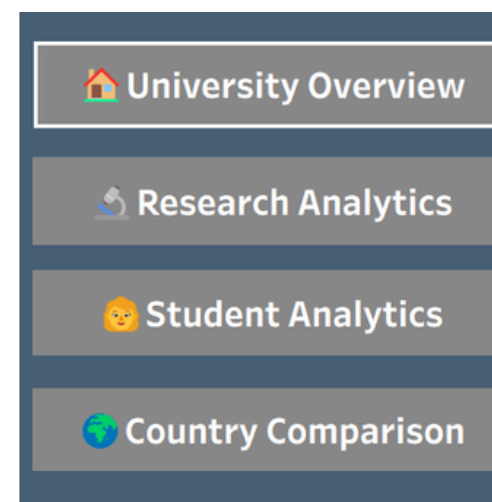
Functionality: A single-click visual button (Reset Filters) placed below the filter panel.

User Benefit: Instantly clears all applied dropdown selections and cross-chart highlights, restoring the dashboard back to its default macro view.



Technical Setup: Targeted dashboard actions [Filter Action]

3. Sidebar Navigation Controls (App-Like Flow) ➔



User Benefit: Replaces default Tableau worksheet tabs with an intuitive, web-app style sidebar menu for seamless switching.

Technical Setup: Navigation buttons were configured using Tableau Dashboard Navigation Actions

4. Parameter Actions & Cross-Dashboard Interactivity



Cross-Highlighting & Tooltips: Clicking on a specific bar, scatter plot data point, or treemap box automatically highlights related attributes.

Dashboard Linking: Inter-dashboard filter passing ensures that selecting a country or institution in Dashboard 1 carries the context over when navigating to Dashboard 2 or 3.

Testing and Validation

Testing Activities



KPI Validation

Verified calculated KPIs against the source data.



Ranking Validation

Checked university and country rankings for calculation accuracy.



Interaction Testing

Tested filters, navigation buttons, and dashboard interactions.



Metric Validation

Cross-checked educational metrics to ensure consistency and correctness.

Project Validation



Dataset Completeness



>95% target achieved



Missing Data



<2% target



KPIs Generated

6+ KPIs generated and validated



Dashboards Integrated

4 dashboards successfully integrated

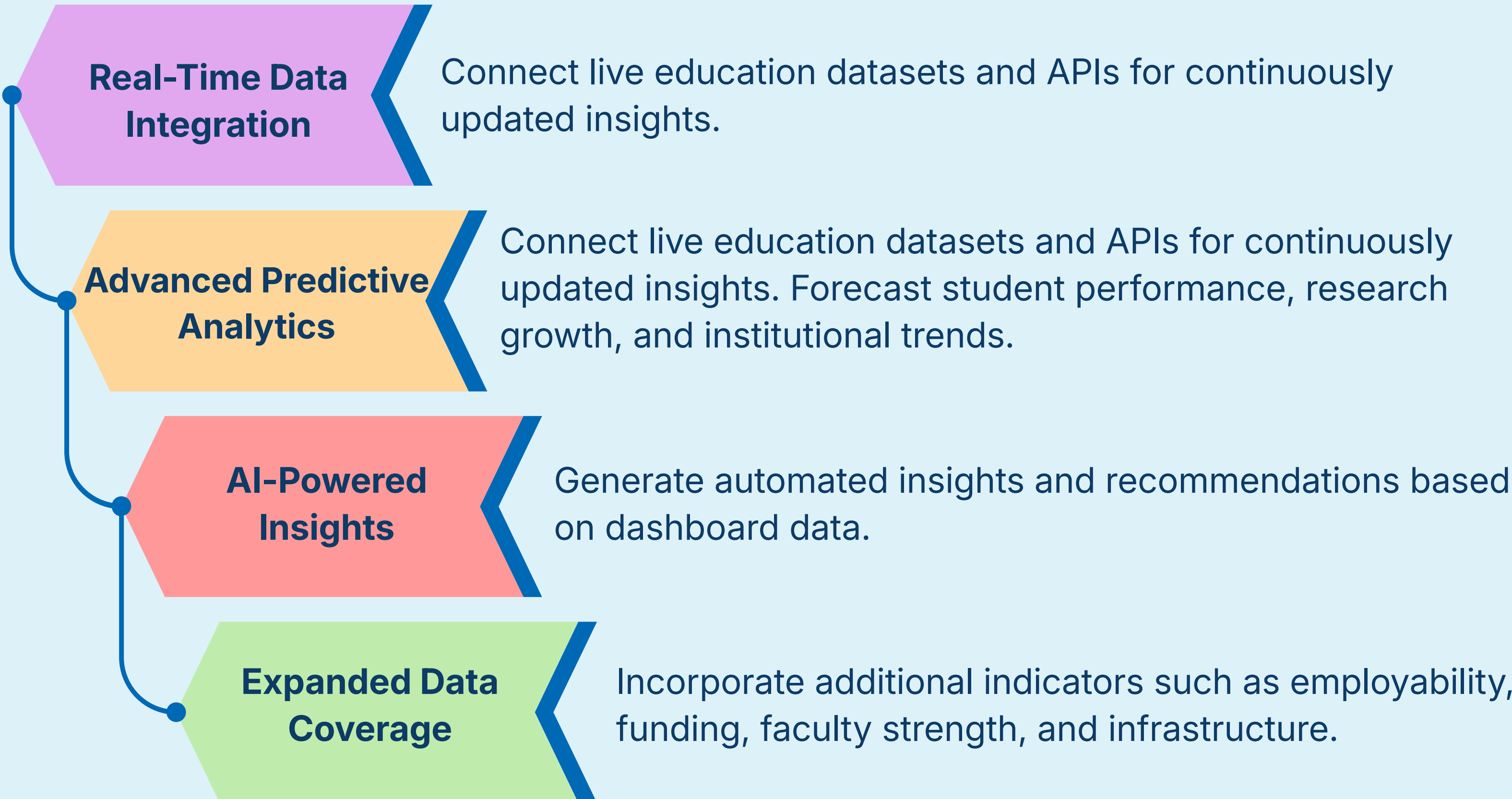


Final Delivery

Fully documented and portfolio-ready dashboard suite



Future Enhancement



Real-Time Data Integration

Connect live education datasets and APIs for continuously updated insights.

Advanced Predictive Analytics

Connect live education datasets and APIs for continuously updated insights. Forecast student performance, research growth, and institutional trends.

AI-Powered Insights

Generate automated insights and recommendations based on dashboard data.

Expanded Data Coverage

Incorporate additional indicators such as employability, funding, faculty strength, and infrastructure.

Conclusion

EduVision transforms higher-education data into clear, interactive, and actionable insights.



DATA TRANSFORMATION

- Simplifies complex educational data.
- Turns raw data into valuable insights.



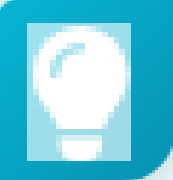
COMPREHENSIVE VIEW

- Dashboards cover universities, students, Research performance, and country-wise trends.



INTERACTIVE EXPLORATION

- Interactive filters and visualizations.
- Enable quick comparison and exploration.



INFORMED DECISION-MAKING

- Supports data-driven analysis and decisions.
- Improves strategic planning.



SCALABLE FOUNDATION

- Provides a basis for deeper analytics.
- Enables future predictive insights.

Thank You